

PROJECT REPORT

1. INTRODUCTION

1.1 Project Overview

This project aims to **Visualizing Housing Market Trends**. The project transforms raw housing data into interactive dashboards that help identify sale price patterns, house age distribution, renovation status, property features, and area statistics. The insights generated support buyers, sellers, and real estate professionals in making data-driven decisions, understanding market trends, and evaluating property values effectively.

1.2 Purpose

The purpose of this project is to analyze housing market trends and property features through interactive data visualization using Tableau. It helps buyers, sellers, and real estate professionals understand sale price patterns, house age distribution, renovation status, and property characteristics. The interactive dashboards support data-driven decision-making by providing meaningful insights into housing market trends and property values.

2. IDEATION PHASE

2.1 Problem Statement

The real estate industry generates large volumes of housing data, including sale prices, property features, house age, and renovation details. However, analyzing raw housing data to identify market trends, price variations, and property characteristics is challenging. This project provides an interactive Tableau dashboard that transforms raw housing data into meaningful visual insights, enabling buyers, sellers, and real estate professionals to make informed, data-driven decisions.

2.2 Empathy Map Canvas

Says

- I want a house at a reasonable price.
- I prefer properties with modern features.
- I look for houses in good locations.

Thinks

- Which property offers the best value?
- Is the sale price worth the property features?
- Has the house been renovated recently?

Does

- Searches available properties.
- Applies filters to compare houses.
- Analyzes house features and sale prices.
- Compares different properties before buying.

Feels

- Happy when finding a suitable property.
- Confident after comparing market trends.
- Satisfied with an informed buying decision.

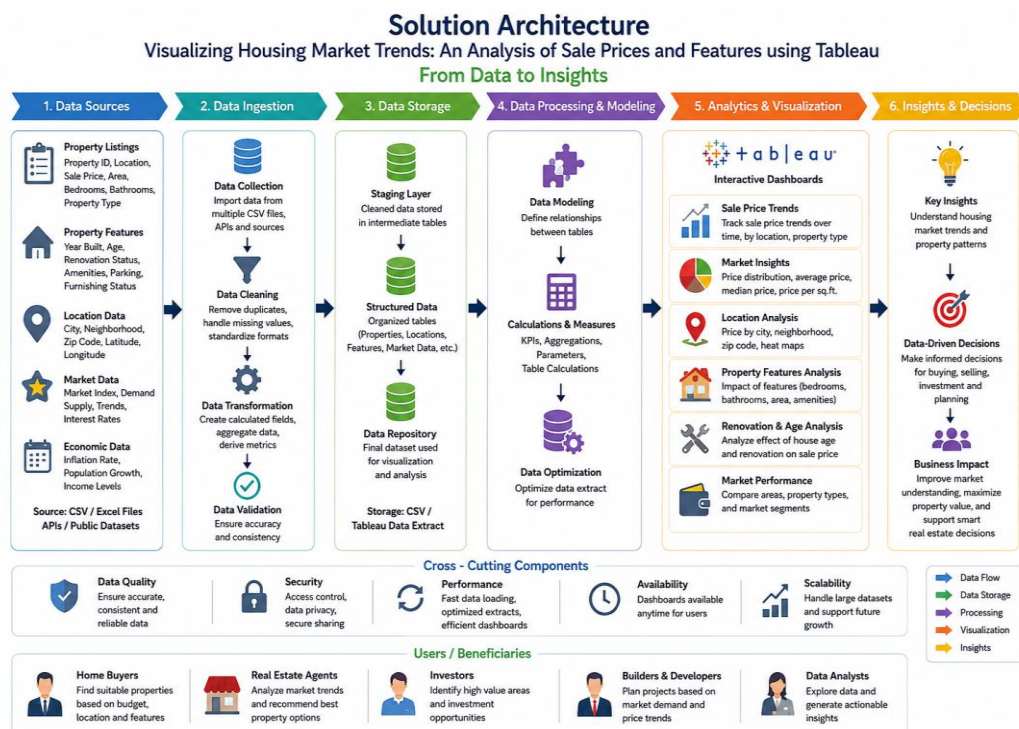
2.3 Brainstorming

- Analyze housing market trends and sale price patterns.
- Compare property features and house characteristics.
- Identify the impact of house age and renovations on sale prices.
- Evaluate house area and property distribution.
- Build interactive Tableau dashboards for visualization.
- Generate valuable insights to support real estate decision-making.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

The user explores housing market data, searches properties based on their requirements, compares sale prices and property features, analyzes house age and renovation status, evaluates different housing options, and makes informed decisions. This journey helps buyers, sellers, and real estate professionals understand market trends and identify the best property opportunities.



3.2 Solution Requirement

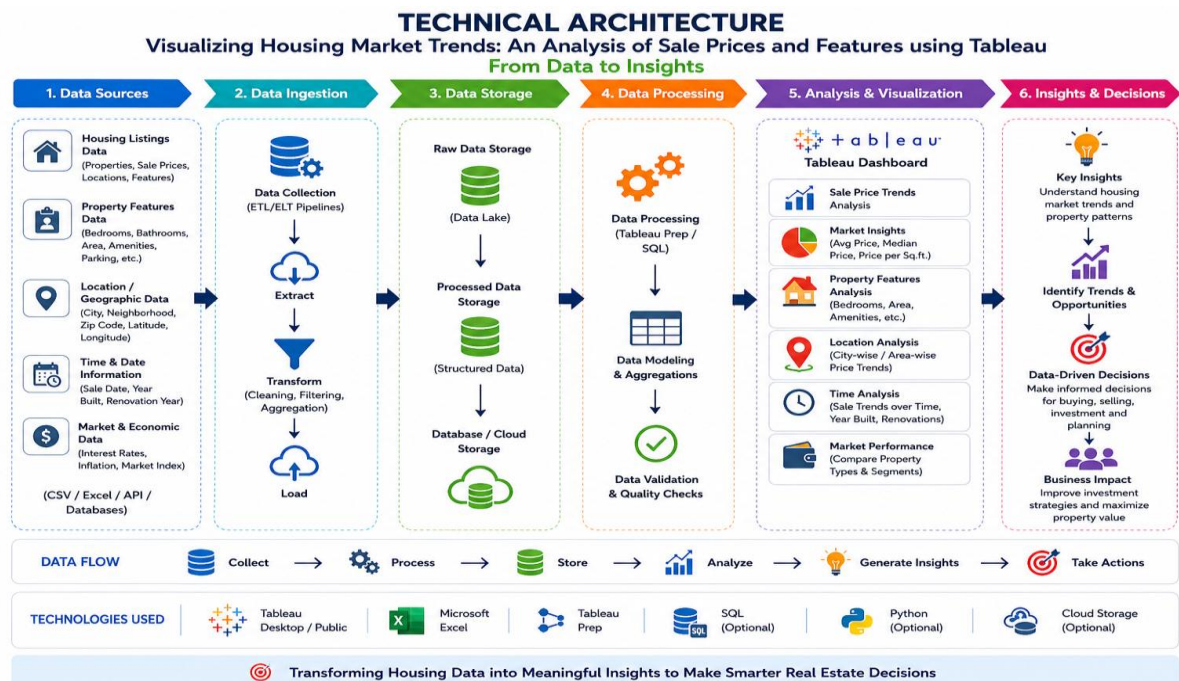
- Interactive Tableau Dashboard
- Data Cleaning and Preparation
- Housing Market Trend Analysis
- Sale Price Analysis
- House Age and Renovation Analysis
- Property Features and Area Analysis
- KPI Dashboard
- Filters and Calculated Fields

3.3 Data Flow Diagram

Housing Dataset (CSV) → Data Cleaning & Preparation → Tableau → Data Analysis & Visualization → Interactive Dashboard → Housing Market Insights

3.4 Technology Stack

- **Tableau Desktop / Tableau Public** – Data Visualization and Dashboard Development
- **CSV Dataset** – Housing Market Data Source
- **Microsoft Excel** – Data Cleaning and Preprocessing
- **Tableau Prep (Optional)** – Data Preparation and Transformation
- **SQL (Optional)** – Data Querying and Analysis



4. PROJECT DESIGN

4.1 Problem Solution Fit

The project transforms raw housing market data into interactive Tableau dashboards, enabling users to analyze sale prices, house age, renovation status, and property features. It provides meaningful insights into housing market trends, helping buyers, sellers, and real estate professionals make informed, data-driven decisions.

4.2 Proposed Solution

Develop an interactive Tableau dashboard that visualizes housing market trends, sale price analysis, house age distribution, renovation status, property features, and location-based insights through interactive charts, KPIs, and filters.

4.3 Solution Architecture

Housing Dataset (CSV) → Data Cleaning & Preparation → Tableau Visualization → Interactive Dashboard → Housing Market Insights

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

The project was executed using the Agile methodology with four sprints covering **housing data collection, data cleaning and preprocessing, Tableau dashboard development, testing, and project documentation.**

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

- Housing dataset loaded successfully.
- Dashboard filters responded correctly.
- Charts and visualizations displayed accurate results.
- Calculated fields and KPIs generated correct values.
- Dashboard performance was smooth and responsive.

7. RESULTS

7.1 Output Screenshots

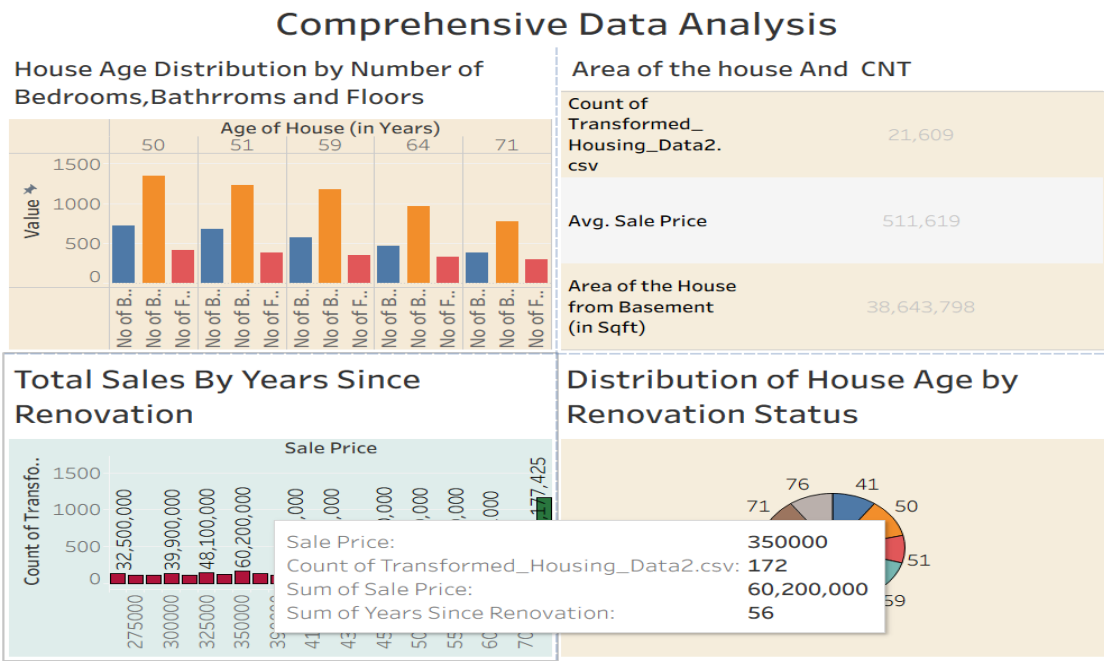
Include screenshots of:

- Data Source
- Worksheets
- Dashboard
- Filters
- Calculated Fields
- Story

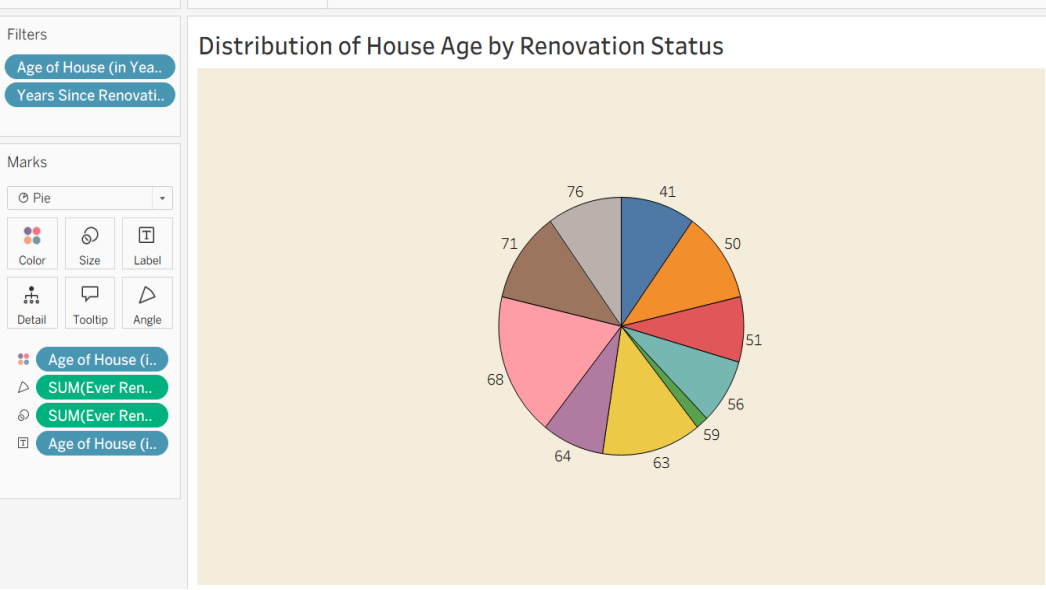
DATA SOURCE

Transformed_Housing_Data2.csv							
10,000 rows 32 fields							
Show Fields							
Transformed_Ho...	#	#		#			
Transformed_Housing_Data2.csv	Transformed_Housing_Data2.csv	Transformed_Housing_Data2.csv	Transformed_Housing_Data2.csv	Transformed_Housing_Data2.csv	Transformed_Housing_Data2.csv	Transformed_Housing_Data2.csv	Transformed_Housing_Data2.csv
Age of House (in Years)	Sale Price	Sale Price (bin)	Years Since Renovation	Zipcode Group Zipcode Gr	Zipcode Group Zipcode Gr	Zipcode Group Zipcode Gr	Zipcode Group Zipcode Gr
63	221900.00	212.994	0	00000	00000	00000	00000
67	538000.00	520.652	26	00000	00001	00000	00000
85	180000.00	165.662	0	00000	00001	00000	00000
53	604000.00	591.650	0	00000	00000	00001	00001
31	510000.00	496.986	0	00000	00000	00000	00000
17	1129575.00	1112.302	0	00000	00000	00000	00000
23	257500.00	236.660	0	00000	00000	00000	00000
55	291850.00	283.992	0	00000	00000	00000	00000
58	229500.00	212.994	0	00001	00000	00000	00000
15	323000.00	307.658	0	00001	00000	00000	00000
53	662500.00	638.982	0	00000	00000	00000	00000
76	468000.00	449.654	0	00000	00000	00000	00000
91	310000.00	307.658	0	00000	00001	00000	00000
41	400000.00	378.656	0	00000	00000	00000	00000
118	530000.00	520.652	0	00000	00000	00001	00001
39	650000.00	638.982	0	00000	00001	00000	00000
24	395000.00	378.656	0	00000	00001	00000	00000
102	485000.00	473.320	0	00000	00000	00001	00001
97	189000.00	165.662	0	00000	00000	00000	00000
49	230000.00	212.994	0	00000	00000	00000	00000
71	385000.00	378.656	0	00001	00000	00000	00000
50	1129575.00	1112.302	0	00000	00000	00000	00000

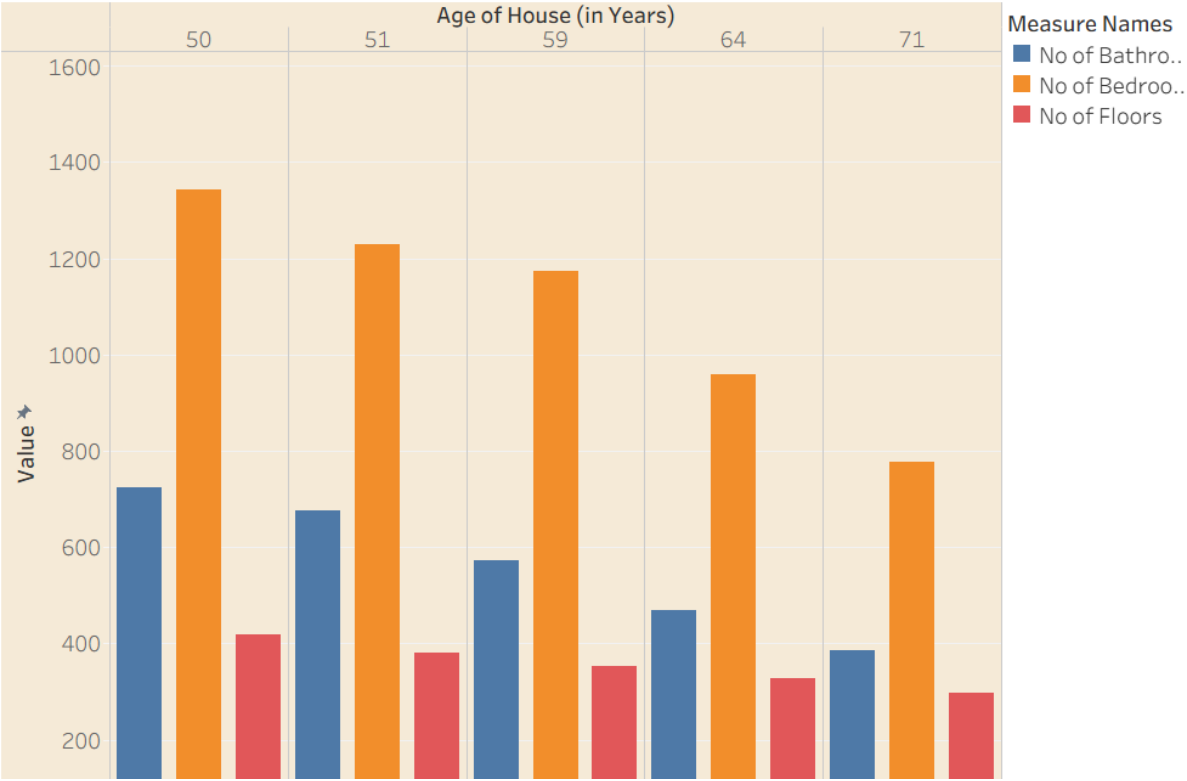
DASHBOARD

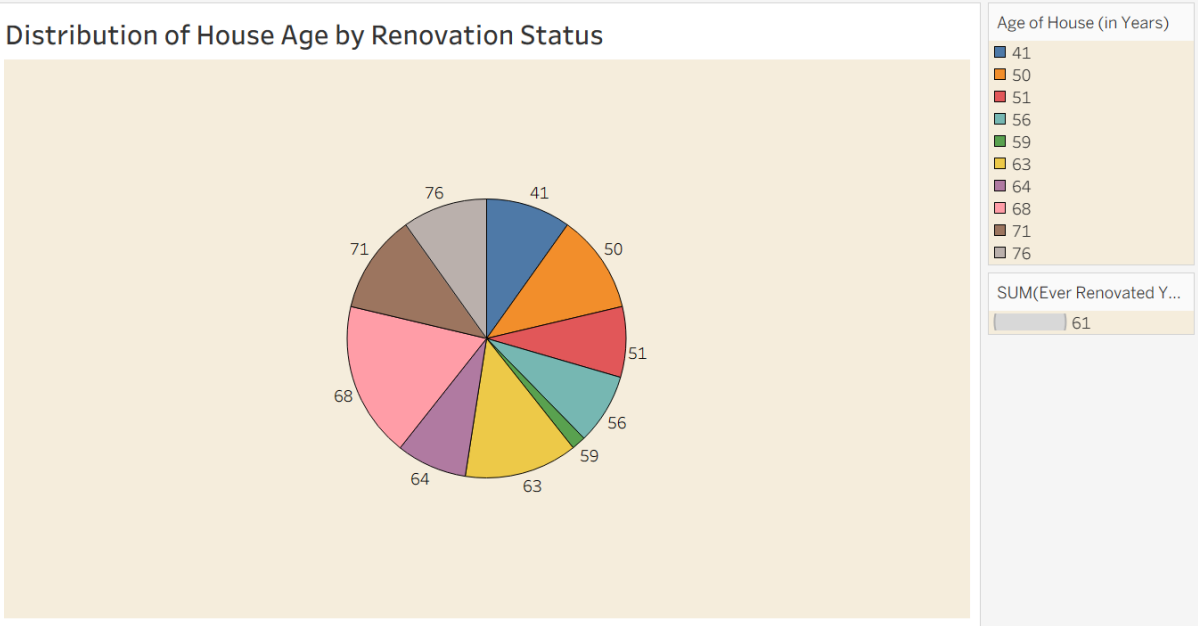
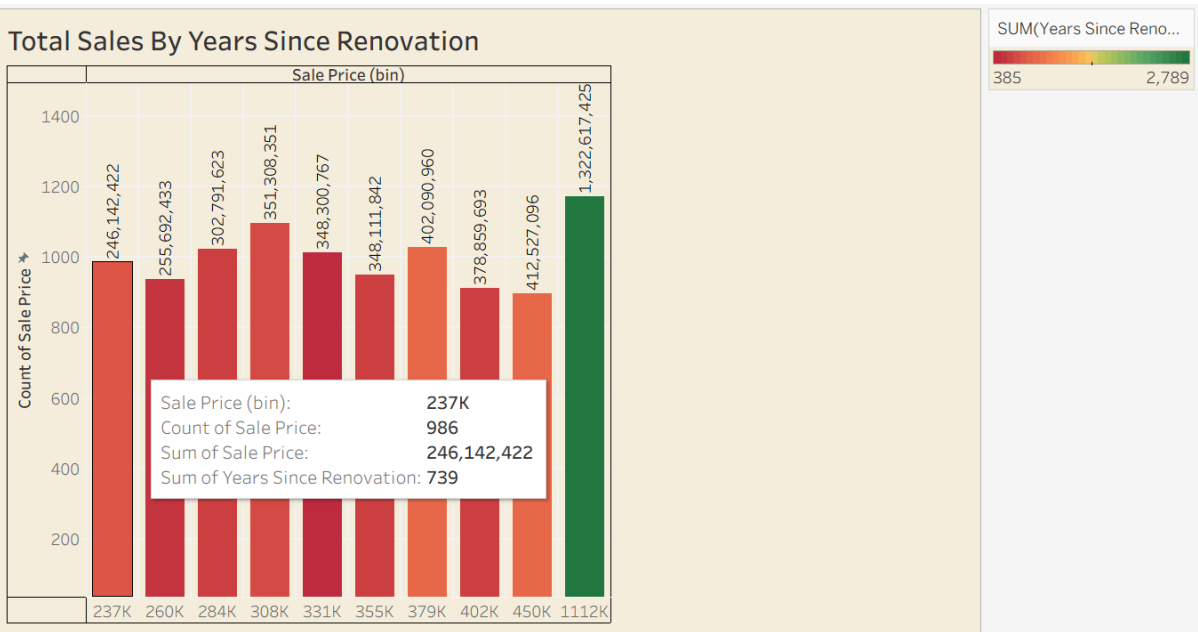


FILTERS



STORY





8. ADVANTAGES & DISADVANTAGES

Advantages

- Interactive housing market dashboards.
- Easy to understand and analyze property data.
- Supports data-driven real estate decisions.
- Helps identify housing market trends quickly.

- Compares sale prices and property features effectively.
- Saves time in housing market reporting and analysis.

Disadvantages

- Depends on the quality of housing data.
- Based only on historical housing records.
- Real-time market analysis is not implemented.
- Requires Tableau knowledge to update or modify dashboards.

9. CONCLUSION

The project successfully analyzes housing market trends and property features using Tableau dashboards. It provides valuable insights into sale prices, house age, renovation status, property area, and market trends, enabling buyers, sellers, and real estate professionals to make informed decisions and improve real estate analysis.

10. FUTURE SCOPE

- Real-time housing market data integration.
- AI-based house price prediction.
- Predictive analytics using Machine Learning.
- Mobile-friendly Tableau dashboards.
- Cloud database integration.
- Location-wise housing market analysis.
- Live real estate market monitoring dashboards.

11. APPENDIX

DATASET LINK

https://drive.google.com/file/d/1ShI7S6_PSPzzyWTJB2iBKyp1lQCl4SD/view?usp=drive_link

GITHUB LINK

PROJECT DEMO LINK

